Integration by Parts

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The integration by parts formula**

State the integration by parts formula, and explain the LIATE guideline for choosing u.

$\int u\,dv = uv - \int v\,du$. LIATE suggests choosing u in the priority order: Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — the function type listed first should typically be u.

2 **[Foundational]** • **Target: The integration by parts formula**

Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.

Integrating both sides: $uv = \int u'v\,dx + \int uv'\,dx = \int v\,du + \int u\,dv$. Rearranging: $\int u\,dv = uv - \int v\,du$.

3 **[Foundational]** • **Target: Repeated integration by parts**

Evaluate $\int x^2 e^x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = e^x\,dx$: $\int x^2 e^x\,dx = x^2e^x - 2\int xe^x\,dx$. Using the earlier result $\int xe^x\,dx = xe^x - e^x$: $= x^2e^x - 2(xe^x - e^x) + C = x^2e^x - 2xe^x + 2e^x + C$.

4 **[Foundational]** • **Target: Repeated integration by parts**

Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x\,dx$, identifying how many times integration by parts must be applied.

Each application of IBP with $u = x^n$ reduces the power of x by 1 while v stays an antiderivative of e^x. Since the power starts at 3, integration by parts must be applied 3 times to eliminate the polynomial factor entirely.

5 **[Foundational]** • **Target: The cyclic case**

Evaluate $\int e^x \sin x\,dx$ using the cyclic (solve-for-the-integral) technique.

Let $I = \int e^x \sin x\,dx$. First IBP ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x\,dx$. Second IBP on the remaining integral ($u = \cos x$, $dv = e^x dx$): $\int e^x \cos x\,dx = e^x \cos x + \int e^x \sin x\,dx = e^x \cos x + I$. So

$I = e^x \sin x - (e^x \cos x + I) \Rightarrow 2I = e^x \sin x - e^x \cos x \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$.

6 **[Foundational]** • **Target: The cyclic case**

Evaluate $\int e^{2x} \cos x\,dx$ using the cyclic technique.

Let $I = \int e^{2x} \cos x\,dx$. IBP ($u = \cos x$, $dv = e^{2x} dx$): $I = \frac{1}{2}e^{2x}\cos x + \frac{1}{2}\int e^{2x}\sin x\,dx$. IBP again on that integral ($u = \sin x$, $dv = e^{2x} dx$): $\int e^{2x}\sin x\,dx = \frac{1}{2}e^{2x}\sin x - \frac{1}{2}I$. Substituting:

$I = \frac{1}{2}e^{2x}\cos x + \frac{1}{4}e^{2x}\sin x - \frac{1}{4}I \Rightarrow \frac{5}{4}I = \frac{1}{2}e^{2x}\cos x + \frac{1}{4}e^{2x}\sin x \Rightarrow I = \frac{e^{2x}(2\cos x + \sin x)}{5} + C$.

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7 [Foundational] • Target: Definite integrals using integration by parts

Evaluate $\int_0^1 xe^x\,dx$ using the antiderivative found earlier.

$[xe^x - e^x]_0^1 = (e - e) - (0 - 1) = 0 - (-1) = 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration by parts to derive a reduction formula

Use integration by parts to show $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x\,dx$.

Let $u = x^n$, $dv = e^x\,dx$: $du = nx^{n-1}dx$, $v = e^x$. $\int x^n e^x\,dx = x^n e^x - n\int x^{n-1}e^x\,dx$, confirming the reduction formula. With $n = 2$: $\int x^2 e^x\,dx = x^2 e^x - 2\int xe^x\,dx = x^2 e^x - 2xe^x + 2e^x + C$, matching the earlier result.

9 [Intermediate] • Target: Choosing u and dv wisely

For each integral, state the best choice of u (using LIATE) without evaluating:

a $\int x^3 \ln x\,dx$ $u = \ln x$ (Logarithmic beats Algebraic)

b $\int xe^{5x}\,dx$ $u = x$ (Algebraic beats Exponential)

c $\int x\tan^{-1}x\,dx$ $u = \tan^{-1}x$ (Inverse trig beats Algebraic)

d $\int x^2\cos x\,dx$ $u = x^2$ (Algebraic beats Trigonometric)

(a) $u = \ln x$ (Logarithmic beats Algebraic); (b) $u = x$ (Algebraic beats Exponential); (c) $u = \tan^{-1}x$ (Inverse trig beats Algebraic); (d) $u = x^2$ (Algebraic beats Trigonometric)

10 [Intermediate] • Target: Choosing u and dv wisely

Explain what goes wrong if you attempt $\int xe^x\,dx$ with the reversed choice $u = e^x$, $dv = x\,dx$.

Then $du = e^x\,dx$, $v = \frac{x^2}{2}$, giving $\int xe^x\,dx = \frac{x^2}{2}e^x - \int \frac{x^2}{2}e^x\,dx$. The remaining integral is MORE complicated than the original (higher power of x under the same exponential), so this choice makes the problem worse rather than simpler — LIATE's algebraic-before-exponential guidance avoids this trap.

11 [Intermediate] • Target: Definite integrals: area under a transcendental product

Find the exact area under $y = xe^{-x}$ on $[0, 3]$, using $\int xe^{-x}\,dx = -xe^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).

Check: $\frac{d}{dx}[-xe^{-x} - e^{-x}] = -e^{-x} + xe^{-x} + e^{-x} = xe^{-x}$ ✓. Area $= [-xe^{-x} - e^{-x}]_0^3 = (-3e^{-3} - e^{-3}) - (0 - 1) = 1 - 4e^{-3} \approx 1 - 0.199 \approx 0.801$.

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12 **[Intermediate]** • **Target: Mixed review: choosing the right technique**

For each integral, state whether it should be solved by u-substitution or integration by parts, and identify the key choice (substitution variable or u) without fully evaluating:

a $\int 2xe^{x^2}\,dx$ Substitution ($w = x^2$) — the derivative of the exponent is present as a factor

b $\int x^2 e^x\,dx$ Integration by parts ($u = x^2$, twice) — no matching derivative factor for substitution

c $\int \frac{\ln x}{x}\,dx$ Substitution ($w = \ln x$) — $\frac{1}{x}$ is exactly dw

d $\int x \ln x\,dx$ Integration by parts ($u = \ln x$) — no simplifying substitution exists here

(a) Substitution ($w = x^2$) — the derivative of the exponent is present as a factor; (b) Integration by parts ($u = x^2$, twice) — no matching derivative factor for substitution; (c) Substitution ($w = \ln x$) — $\frac{1}{x}$ is exactly dw; (d) Integration by parts ($u = \ln x$) — no simplifying substitution exists here

13 **[Intermediate]** • **Target: Repeated integration by parts**

Level 3 · Advanced / AP FRQWrite the definite integral that represents the total perimeter (arc length) of the cardioid $r = 1 + \cos\theta$, and simplify the integrand algebraically into a single trigonometric function.

Compute $r'(\theta) = -\sin\theta$. Form the radicand: $r^2 + (r')^2 = (1+\cos\theta)^2 + (-\sin\theta)^2 = 1 + 2\cos\theta + \cos^2\theta + \sin^2\theta = 2 + 2\cos\theta = 2(1+\cos\theta)$. Recall the half-angle identity $1 + \cos\theta = 2\cos^2(\theta/2)$: $r^2 + (r')^2 = 2\left(2\cos^2(\theta/2)\right) = 4\cos^2(\theta/2)$. Taking the square root for $\theta \in [0, \pi]$: $\sqrt{4\cos^2(\theta/2)} = 2\cos(\theta/2)$. By symmetry across $[0, 2\pi]$.

14 **[Intermediate]** • **Target: The cyclic case**

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x\,dx$.

1. First IBP: Let $u = x^2 \Rightarrow du = 2x\,dx$. Let $dv = \cos x\,dx \Rightarrow v = \sin x$. $\int_0^{\pi/2} x^2\cos x\,dx = [x^2\sin x]_0^{\pi/2} - 2\int_0^{\pi/2} x\sin x\,dx = \left(\left(\frac{\pi}{2}\right)^2(1) - 0\right) - 2\int_0^{\pi/2} x\sin x\,dx = \frac{\pi^2}{4} - 2J.$ 2. Second IBP on $J = \int_0^{\pi/2} x\sin x\,dx$: Let $u = x \Rightarrow du = dx$. Let $dv = \sin x\,dx \Rightarrow v = -\cos x$..

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Definite integrals using integration by parts** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Level 2 · MediumEvaluate $\int x^3 \sin(2x)\,dx$ using the Tabular Method.

Tabulate signs, derivatives of x^3, and antiderivatives of $\sin(2x)$: • $(+)$ x^3 $-\frac{1}{2}\cos(2x)$ • $(-)$ $3x^2$ $-\frac{1}{4}\sin(2x)$ • $(+)$ $6x$ $\frac{1}{8}\cos(2x)$ • $(-)$ 6 $\frac{1}{16}\sin(2x)$ • $(+)$ 0 $-\frac{1}{32}\cos(2x)$ Sum the diagonal products: $= -\frac{1}{2}x^3\cos(2x) + \frac{3}{4}x^2\sin(2x) + \frac{3}{4}x\cos(2x) - \frac{3}{8}\sin(2x) + C.$ Group terms.

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16 [Challenging] • Target: Integration by parts to derive a reduction formula • Cross-Lesson: 4.02 Area in Polar Coordinates

Level 3 · Advanced / AP FRQA particle moves along a planar path with velocity $\vec{v}(t) = \langle \sin(t^2), \cos(t^3) \rangle$ for $0 \le t \le 3$. Write the definite integral representing the total distance traveled by the particle, and calculate the numerical value using three decimal places.

Written Setup: Total Distance $= \int_0^3 \sqrt{\sin^2(t^2) + \cos^2(t^3)}\, dt$. Numerical Evaluation: Entering this into a graphing calculator yields: Total Distance $\approx \mathbf{2.874}$ (or 2.873).

17 [Challenging] • Target: Choosing u and dv wisely • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

[Unit 6BC · Advanced Integration Techniques · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 [Challenging] • Target: Integration by parts with algebraic-trigonometric products • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

[Unit 9 · Polar Coordinates & Polar Area · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Definite integrals: area under a transcendental product • Cross-Lesson: 6.01 Power Series and Interval of Convergence

(Paper 3 Extended Investigation) Define the integral family $I_n = \int_0^1 x^n e^x\, dx$ for integers $n \ge 0$. (a) Evaluate I_0. (b) Use integration by parts to establish the reduction formula $I_n = e - nI_{n-1}$ for $n \ge 1$. (c) Hence determine the exact value of $\int_0^1 x^3 e^x\, dx$. (d) Prove that $0 < I_n < \frac{e}{n+1}$, and deduce $\lim_{n\to\infty} I_n$.

(a) Base integral: $I_0 = \int_0^1 e^x\, dx = [e^x]_0^1 = e^1 - e^0 = e - 1$. (b) Reduction formula via IBP: Let $u = x^n \Rightarrow du = nx^{n-1}\, dx$. Let $dv = e^x\, dx \Rightarrow v = e^x$. $I_n = [x^n e^x]_0^1 - n\int_0^1 x^{n-1} e^x\, dx = (1^n e^1 - 0) - nI_{n-1} = e - nI_{n-1}$. ■ (c) Compute I_3: - $I_1 = e - 1(I_0) = e - (e-1) = 1$. - $I_2 = e - 2(I_1) = e - 2(1) = e - 2$. - $I_3 = e - 3(I_2) = e - 3(e-2) = e - 3e + 6 = 6 - 2e$. (d) Squeeze Theorem & Limit: For.

20 [Challenging] • Target: Mixed review: choosing the right technique • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Level 3 · AdvancedLet $h(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. Prove that h is strictly increasing on $\mathbb{R}$ without using calculus derivatives, deduce that h is injective, and derive the explicit formula for $h^{-1}(x)$ including its domain.

Step 1 · Algebraic Monotonicity: Rewrite $h(x)$ by dividing numerator and denominator by e^x: $h(x) = \frac{1 - e^{-2x}}{1 + e^{-2x}} = \frac{(1 + e^{-2x}) - 2e^{-2x}}{1 + e^{-2x}} = 1 - \frac{2}{e^{2x} + 1}$. As x, $\frac{2}{e^{2x}+1} \in (0, 2) \Rightarrow h(x) = 1 - \frac{2}{e^{2x}+1} \in (-1, 1)$. The domain of h^{-1} is $(-1, 1)$. Set $y = \frac{e^{2x} - 1}{e^{2x} + 1}$: $y(e^{2x} + 1) = e^{2x} - 1 \Rightarrow ye^{2x} + y = e^{2x} - 1 \Rightarrow e^{2x}(1 - y) = 1 + y$. For $y \in (-1, 1)$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The integration by parts formula

Evaluate $\int_0^{\pi} -2x\cos x\,dx$, using integration by parts.

a. 2

b. 6

c. 4 ✓

d. -4

Let $u=-2x$, $dv=\cos x\,dx$: $du=-2\,dx$, $v=\sin x$. $\int -2x\cos x\,dx=-2x\sin x-2\cos x+C$.
Evaluating: $[-2x\sin x-2\cos x]_0^{\pi}=(0+2)-(0-2)=4$.

22 [Foundational] • Target: Integration by parts with logarithms and inverse trig

Given $f(0)=0$, $f(3)=2$, $g(0)=2$, $g(3)=2$, and $\int_0^3 f'(x)g(x)\,dx=-2$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. 10

b. -6

c. 2

d. 6 ✓

$\int_0^3 f(x)g'(x)\,dx=[f(x)g(x)]_0^3-\int_0^3 f'(x)g(x)\,dx=[(2)(2)-(0)(2)]-(-2)=4-(-2)=6$.

23 [Foundational] • Target: Repeated integration by parts

Using the cyclic technique, $\int e^x\cos(2x)\,dx=\frac{e^x(\cos(2x)+C_1\sin(2x))}{5}+C$. Find the coefficient C_1.

a. $\frac{1}{5}$

b. $-\frac{2}{5}$

c. 2

d. $\frac{2}{5}$ ✓

By the standard cyclic-IBP result, $\int e^{ax}\cos(bx)\,dx=\frac{e^{ax}(a\cos(bx)+b\sin(bx))}{a^2+b^2}+C$. With $a=1$, $b=2$: $C_1=b=2$, and the denominator is $1^2+2^2=5$, so as a fraction of the denominator, $C_1=\frac{2}{5}=\frac{2}{5}$.

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24 [Foundational] • Target: The cyclic case

For $\int x \ln x \, dx$, which is the correct choice of u according to LIATE?

a. It does not matter which factor is chosen for u

b. $u = x$ (Algebraic always goes first)

c. $u = x$ (the opposite LIATE priority)

d. $u = \ln x$ **(Logarithmic outranks Algebraic in LIATE)** ✓

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u: $u = \ln x$ (Logarithmic outranks Algebraic in LIATE).

25 [Foundational] • Target: Definite integrals using integration by parts

Consider a region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the y-axis. Let S be the solid generated when R is revolved about the y-axis. The volume of S can be calculated using the method of cylindrical shells or the disk/washer method. However, direct application of standard integration techniques to find the volume of S leads to an integral that cannot be evaluated in terms of elementary functions. A related problem involves finding the surface area of a solid of revolution. Let C be the curve defined by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. This curve represents a quarter-circle. If this curve C is revolved about the x-axis, it generates a hemisphere. The surface area of this hemisphere (excluding the base) is given by the integral $A = \int_0^1 2\pi y \sqrt{1 + (y')^2} \, dx$. Determine the exact surface area of the generated hemisphere.

a. $\frac{\pi}{2}$

b. π ✓

c. 2π

d. 4π

The curve is given by $y = \sqrt{1 - x^2}$ for $0 \le x \le 1$. We first find the derivative of y with respect to x: $y' = \frac{d}{dx}(\sqrt{1 - x^2}) = \frac{1}{2\sqrt{1 - x^2}} \cdot (-2x) = \frac{-x}{\sqrt{1 - x^2}}$. Next, we compute $1 + (y')^2$: $1 + (y')^2 = 1 + \left(\frac{-x}{\sqrt{1 - x^2}}\right)^2 = 1 + \frac{x^2}{1 - x^2} = \frac{1 - x^2 + x^2}{1 - x^2} = \frac{1}{1 - x^2}$. Then, $\sqrt{1 + (y')^2} = \sqrt{\frac{1}{1 - x^2}} = \frac{1}{\sqrt{1 - x^2}}$ for $0 \le x < 1$. The surface area integral is.

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26 [Foundational] • Target: Integration by parts to derive a reduction formula

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. What is the definite integral of $f(x)$ from $x = 1$ to $x = 3$?

a. $-\frac{4}{3}$ ✓

b. 0

c. $\frac{4}{3}$

d. 2

To find the definite integral of $f(x) = x^2 - 4x + 3$ from $x = 1$ to $x = 3$, we evaluate $\int_1^3 (x^2 - 4x + 3)\,dx$. First, find the antiderivative of $f(x)$: $F(x) = \frac{x^3}{3} - \frac{4x^2}{2} + 3x = \frac{x^3}{3} - 2x^2 + 3x$. Next, apply the Fundamental Theorem of Calculus: $F(3) - F(1)$. Calculate $F(3)$: $F(3) = \frac{3^3}{3} - 2(3)^2 + 3(3) = \frac{27}{3} - 2(9) + 9 = 9 - 18 + 9 = 0$. Calculate $F(1)$: $F(1) = \frac{1^3}{3} - 2(1)^2 + 3(1) = \frac{1}{3} - 2 + 3 = \frac{1}{3} + 1 = \frac{4}{3}$. Fi.

27 [Foundational] • Target: Choosing u and dv wisely

Consider the region bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?

a. $\sqrt{2}$

b. $\frac{1}{\sqrt{3}}$

c. $\sqrt{3}$ ✓

d. 2

The area of the region is given by the definite integral of the function from $x = 0$ to $x = b$. The integral of $\frac{1}{x^2 + 1}$ is $\arctan(x)$. Therefore, the area is given by: $\text{Area} = \int_0^b \frac{1}{x^2+1} dx = [\arctan(x)]_0^b = \arctan(b) - \arctan(0) = \arctan(b)$ We are given that the area is $\frac{\pi}{3}$. So, we have: $\arctan(b) = \frac{\pi}{3}$ To find b, we take the tangent of both sides: $b = \tan\left(\frac{\pi}{3}\right)$ We know that.

28 [Foundational] • Target: Integration by parts with algebraic-trigonometric products

A region in the first quadrant is bounded by the graph of $f(x) = 3x^2 + 1$, the x-axis, and the vertical lines $x = 1$ and $x = 2$. What is the area of this region?

a. 10 ✓

b. 12

c. 14

d. 16

The area of the region is given by the definite integral of $f(x)$ from $x = 1$ to $x = 2$. Area $= \int_1^2 (3x^2 + 1)\,dx$ First, find the antiderivative of $3x^2 + 1$: Antiderivative $= \frac{3x^3}{3} + x = x^3 + x$ Now, evaluate the definite integral using the Fundamental Theorem of Calculus: Area $= [x^3 + x]_1^2$ Area $= (2^3 + 2) - (1^3 + 1)$ Area $= (8 + 2) - (1 + 1)$ Area $= 10 - 2$ Area $= 8$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Definite integrals: area under a transcendental product**

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

a. $\frac{5\pi}{16}$

b. $\frac{5\pi}{32}$ ✓

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

On $[0, \pi/2]$, $I_n = \frac{n-1}{n} I_{n-2}$ with $I_0 = \pi/2$ and $I_1 = 1$. Applying it repeatedly gives $\frac{5\pi}{32}$.

30 **[Intermediate]** • **Target: Mixed review: choosing the right technique**

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?

a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$ ✓

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

The region R is bounded above by $y = 4$ and below by $y = x^2$. To find the limits for x, we find the intersection points of the two curves: $x^2 = 4 \Rightarrow x = \pm 2$. Thus, the region is defined for x from -2 to 2. When integrating with respect to y first ($dydx$), the inner integral goes from the lower curve to the upper curve, so from $y = x^2$ to $y = 4$. The outer integral is over the range of x, which is -2 to 2. Therefore, the double integral representing the area is.

31 **[Intermediate]** • **Target: The integration by parts formula**

Consider the function represented by the graph of $f(x)$ below. What is the value of the definite integral $\int_0^4 f(x)\,dx$?

a. **12** ✓

b. 10

c. 8

d. 6

The definite integral $\int_0^4 f(x)\,dx$ represents the area under the curve $f(x)$ from $x = 0$ to $x = 4$. The graph of $f(x)$ from $x = 0$ to $x = 4$ forms a triangle with a base of 4 units (from 0 to 4 on the x-axis) and a height of 6 units (the value of $f(x)$ at $x = 0$). The area of a triangle is given by $\frac{1}{2} \times \text{base} \times \text{height}$. Therefore, the area is $\frac{1}{2} \times 4 \times 6 = 12$.

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32 [Intermediate] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int x^2 e^{2x}\,dx$ (apply integration by parts twice).

a. $\left(\frac{x^2}{2}+\frac{2x}{4}+\frac{2}{8}\right)e^{2x}+C$

b. $\left(\frac{x^2}{2}-\frac{2x}{4}+\frac{2}{8}\right)e^{2x}+C$ ✓

c. $\left(\frac{x^2}{2}-\frac{2x}{4}\right)e^{2x}+C$

d. $x^2e^{2x}-2xe^{2x}+2e^{2x}+C$

Apply IBP twice with $u=x^2$, then $u=x$: $\int x^2e^{2x}dx=\frac{x^2}{2}e^{2x}-\frac{2}{2}\int xe^{2x}dx=\left(\frac{x^2}{2}-\frac{2x}{4}+\frac{2}{8}\right)e^{2x}+C$.

33 [Intermediate] • Target: Repeated integration by parts

Integration by parts: $\int u\,dv=$

a. $uv+\int v\,du$

b. $uv-\int v\,du$ ✓

c. $u\cdot v$

d. $\frac{u}{v}$

Applying Unit 6BC · Advanced Integration Techniques principles, IBP formula: $uv-\int v\,du$. Therefore, the verified correct option is $uv-\int v\,du$.

34 [Intermediate] • Target: The cyclic case

For $\frac{1}{(x-2)^2(x-6)}=\frac{A}{x-2}+\frac{B}{(x-2)^2}+\frac{D}{x-6}$, find B (coefficient of the repeated-factor term).

a. $-\frac{1}{2}$

b. $-\frac{1}{4}$ ✓

c. $-\frac{3}{20}$

d. $\frac{1}{4}$

Multiply through by $(x-2)^2(x-6)$ and set $x=2$: $1=B(2-6)\Rightarrow B=\frac{1}{2-6}=-\frac{1}{4}$.

35 [Intermediate] • Target: Definite integrals using integration by parts

Convert the polar point $(4,\ \pi/4)$ to rectangular coordinates.

a. $(2\sqrt{2},\ 2\sqrt{2}+1)$

b. $(2\sqrt{2},\ 2\sqrt{2})$ ✓

c. $(-2\sqrt{2},\ 2\sqrt{2})$

d. $(4,\ \pi/4)$

$x=r\cos\theta=4\cos(\pi/4)=2\sqrt{2}$, $y=r\sin\theta=4\sin(\pi/4)=2\sqrt{2}$.

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36 **[Intermediate]** • **Target: Integration by parts to derive a reduction formula**

A point is represented in polar coordinates by $(r, \theta) = (5, \frac{\pi}{4})$. What are the equivalent Cartesian coordinates (x, y)?

a. $(\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2})$ ✓

b. $(5, 5)$

c. $(\frac{5}{2}, \frac{5}{2})$

d. $(\frac{5}{2}, \frac{5\sqrt{3}}{2})$

The conversion formulas from polar coordinates (r, θ) to Cartesian coordinates (x, y) are $x = r\cos(\theta)$ and $y = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{4}$, we have: $x = 5\cos(\frac{\pi}{4}) = 5 \cdot \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}$ $y = 5\sin(\frac{\pi}{4}) = 5 \cdot \frac{\sqrt{2}}{2} = \frac{5\sqrt{2}}{2}$ Therefore, the Cartesian coordinates are $(\frac{5\sqrt{2}}{2}, \frac{5\sqrt{2}}{2})$.

37 **[Intermediate]** • **Target: Choosing u and dv wisely**

$\int \sin^4 x \, dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$ ✓

c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x \cos x}{4} + \frac{\sin x \cos x}{4} + \frac{x}{8} + C$

Double application of half-angle.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging]** • **Target: Integration by parts with algebraic-trigonometric products** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Consider the region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = a$, where $a > 0$. A solid is generated by revolving this region R about the y-axis. If the volume of this solid is $\frac{\pi}{2}$, determine the value of a.

a. $\ln(2)$

b. $\sqrt{\ln(2)}$ ✓

c. $e^{\sqrt{2}}$

d. $\sqrt{2}$

The volume of the solid generated by revolving the region bounded by $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ about the y-axis can be found using the shell method. The formula for the volume V is given by $V = \int_a^b 2\pi x f(x) dx$. In this problem, $f(x) = e^{-x^2}$, and the region is bounded by $y = e^{-x^2}$, the x-axis, and the line $x = a$ in the first quadrant. Thus, the limits of integration are from $x = 0$ to $x = a$. The volume of the solid is given by.

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39 [Challenging] • Target: Definite integrals: area under a transcendental product • Cross-Lesson: 6.01 Power Series and Interval of Convergence

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1]. ✓

(a) When revolving the region R about the line $y = -2$, the outer radius $R(x)$ is the distance from $y = -2$ to the upper boundary of the region $y = e^{-x^2}$, which is $R(x) = e^{-x^2} - (-2) = e^{-x^2} + 2$. The inner radius $r(x)$ is the distance from $y = -2$ to the lower boundary of the region (the x-axis, $y = 0$), which is $r(x) = 0 - (-2) = 2$. (b) Using the washer method, the volume V is given by the integral of the area of the washers, which is $\pi(R(x)^2 - r(x)^2)$. Therefore, the integral express.

40 [Challenging] • Target: Mixed review: choosing the right technique • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A cardioid is described by the polar equation $r = a + b\sin(\theta)$, where a and b are positive constants. The graph of this cardioid passes through the point $(r, \theta) = (5, \frac{\pi}{2})$ and the point $(r, \theta) = (3, \frac{3\pi}{2})$. What is the value of the parameter a?

a. 2

b. 3

c. 4 ✓

d. 5

The polar equation of the cardioid is given by $r = a + b\sin(\theta)$. We are given two points that lie on the cardioid: $(r, \theta) = (5, \frac{\pi}{2})$ and $(r, \theta) = (3, \frac{3\pi}{2})$. We can substitute these points into the equation to form a system of two linear equations with two unknowns, a and b. For the first point, $r = 5$ and $\theta = \frac{\pi}{2}$: $5 = a + b\sin(\frac{\pi}{2})$ Since $\sin(\frac{\pi}{2}) = 1$, this simplifies to: $5 = a + b(1)$ Equation 1: $a + b = 5$ For the second p.

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41 [Challenging] • Target: The integration by parts formula • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The rate of change of a quantity $Q(t)$ is given by $\frac{dQ}{dt} = kte^{-0.5t^2}$, where k is a positive constant. If $Q(0) = 5$ and $Q(2) = 15$, find the value of k.

a. $\frac{5}{1-e^{-2}}$

b. $\frac{5}{e^{-2}-1}$

c. $\frac{10}{1-e^{-2}}$ ✓

d. $\frac{10}{e^{-2}-1}$

We are given the differential equation $\frac{dQ}{dt} = kte^{-0.5t^2}$. To find $Q(t)$, we need to integrate both sides with respect to t. The integral of the right side requires a substitution. Let $u = -0.5t^2$, so $du = -t\,dt$ or $-du = t\,dt$. Thus, $kte^{-0.5t^2}\,dt = -ke^u\,du$. Integrating $\int kte^{-0.5t^2}dt$ gives $\int -ke^u\,du = -ke^u + C = -ke^{-0.5t^2} + C$. So, $Q(t) = -ke^{-0.5t^2} + C$. We are given $Q(0) = 5$. Substituting $t = 0$ into the equation for $Q(t)$.

42 [Challenging] • Target: Integration by parts with logarithms and inverse trig • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

What is the area of the region bounded by the polar curve $r = \sqrt{5\sin\theta}$ for $0 \le \theta \le \pi$?

a. 5 ✓

b. 10

c. $\frac{5}{2}$

d. 5π

The area A of a region bounded by a polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$ is given by the formula: $A = \frac{1}{2}\int_{\alpha}^{\beta} r^2\,d\theta$ Substitute $r = \sqrt{5\sin\theta}$, $\alpha = 0$, and $\beta = \pi$: $A = \frac{1}{2}\int_0^{\pi}(\sqrt{5\sin\theta})^2\,d\theta = \frac{1}{2}\int_0^{\pi}5\sin\theta\,d\theta = \frac{5}{2}\int_0^{\pi}\sin\theta\,d\theta$ Evaluate the definite integral.

43 [Challenging] • Target: Repeated integration by parts • Cross-Lesson: 3.01 Parametric Equations

The graph of the function $f(x)$ is shown below. What is the value of the definite integral $\int_{-2}^{2} f(x)\,dx$?

a. 0

b. 2

c. **4** ✓

d. 8

The definite integral $\int_{-2}^{2} f(x)\,dx$ represents the net signed area between the graph of $f(x)$ and the x-axis from $x = -2$ to $x = 2$. The graph of $f(x)$ for $-2 \le x \le 2$ forms a semicircle above the x-axis with radius 2. The area of this semicircle is $\frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2^2) = 2\pi$. However, the provided figure shows a region that is a combination of triangles and a rectangle. From $x = -2$ to $x = 0$, the area is a triangle with base 2 and height 2, so the area is.

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44 [Challenging] • Target: The cyclic case • Cross-Lesson: 3.02 Vector-Valued Functions

A population of bacteria follows a logistic growth model where $P(t)$ is the population at time t (in hours). The differential equation governing the population growth is $\frac{dP}{dt} = 0.05P\left(1 - \frac{P}{1000}\right)$. If the initial population is $P(0) = 200$, use Euler's method with a step size of $\Delta t = 1$ hour to approximate the population after 1 hour, $P(1)$.

a. 210

b. **208 ✓**

c. 212

d. 209

Euler's method approximates the next value $P(t + \Delta t)$ using the formula $P(t + \Delta t) \approx P(t) + \frac{dP}{dt}|_t \cdot \Delta t$. We are given $P(0) = 200$ and $\Delta t = 1$. First, we find the rate of change at $t = 0$: $\frac{dP}{dt}|_{t=0} = 0.05 \cdot P(0)\left(1 - \frac{P(0)}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 0.05 \cdot 200\left(1 - \frac{200}{1000}\right)$ $\frac{dP}{dt}|_{t=0} = 10(1 - 0.2)$ $\frac{dP}{dt}|_{t=0} = 10 \cdot 0.8$ $\frac{dP}{dt}|_{t=0} = 8$ Now, we.

45 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 4.01 Polar Curves and Derivatives

A differentiable function $f(x)$ is represented by the graph below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 1$ to $x = 4$ is given by the definite integral $\int_1^4 f(x)\,dx$. If $f(x) = -x + 5$ for $1 \le x \le 4$, what is the area of this region?

a. 4.5

b. 6

c. **7.5 ✓**

d. 9

The area of the shaded region is given by the definite integral $\int_1^4 (-x + 5)\,dx$. To evaluate this integral, we find the antiderivative of $f(x) = -x + 5$, which is $F(x) = -\frac{x^2}{2} + 5x$. Now we apply the Fundamental Theorem of Calculus: $\int_1^4 (-x + 5)\,dx = F(4) - F(1)$ $F(4) = -\frac{4^2}{2} + 5(4) = -\frac{16}{2} + 20 = -8 + 20 = 12$ $F(1) = -\frac{1^2}{2} + 5(1) = -\frac{1}{2} + 5 = 4.5$ Therefore, the area is $12 - 4.5 = 7.5$. Common errors include incorrectly calculating the anti.

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